Ganesh Singha

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Career Objective

Results-driven Computer Science professional with an MCA from Brainware University (CGPA: 9.01) and hands-on experience building full-stack applications using Java, Spring Boot, and modern web and mobile technologies. Proven ability to deliver production-grade systems — including a patented EV charging platform, an AI-powered IoT chatbot, and a real-time EV station finder app — across backend, Android, and Frontend ecosystems. Currently working as a Junior Engineer at SEPLE NovaEdge Pvt. Ltd., actively expanding expertise in IoT, AI integration, and cloud-based architectures.

Technical Skills

Frontend:	React, Next.js, Flutter, Android SDK (Kotlin), JavaScript, Redux Toolkit, Material UI
Backend:	Java, Spring Boot, Spring Security, Spring MVC, Hibernate, REST API, Maven, MySQL, PostgreSQL, Redis
DevOps:	Git, GitHub, Docker, CI/CD, Firebase, Postman, Maven, AWS
AI Tools:	OpenAI API, Claude Code (Anthropic), CLI
Tools:	IntelliJ IDEA, Android Studio, Visual Studio Code, Antigravity

Education

Brainware University *2024 – 2026*
Master of Computer Applications CGPA: 9.01

MAKAUT *2021 – 2024*
Bachelor of Computer Applications CGPA: 8.43

Experience

Junior Engineer | *SEPLE NovaEdge Pvt. Ltd., Kolkata* *December 2025 – Present*

Collaborated within a three-member engineering team to design and deliver two IoT-based software products on the ThingsBoard PE platform. Built an AI-powered chatbot integrating OpenAI GPT that reduced manual device data querying time by approximately 60% through natural language automation. Contributed to SWatch360, a Flutter cross-platform mobile application used for real-time IoT device monitoring and control.

Projects:

- **ThingsBoard IoT Chatbot** | An AI-powered IoT chatbot integrating ThingsBoard and OpenAI GPT to enable natural language querying of real-time device telemetry and attributes. *[GitHub]*
- **SWatch360 – IoT Mobile Application** | A Flutter-based cross-platform mobile app built on ThingsBoard PE for real-time IoT device monitoring and control. *[GitHub]*

Java Full Stack Intern | *Ardent Computech Pvt. Ltd., Kolkata* *July 2024 – August 2024*

Collaborated in a two-member team to architect and implement a real-time chat web application from the ground up. Owned the Spring Boot backend by creating REST API endpoints and implementing core server-side business logic, accelerating development through reusable service components.

Projects:

- **Chat Web Application** | A real-time web-based chat application with Spring Boot backend and WebSocket support for instant messaging between users. *[GitHub]*

Projects

EV Charging Platform | *Patent No. 202531058380* *February 2026 – Present*

Architected and patented a full-stack EV charging ecosystem that supports Drivers, Station Owners, and Administrators, featuring a Spring Boot REST API backend, a native Kotlin Android app, and a React web dashboard. Enforced real-time slot availability tracking, role-based access control (RBAC), and dynamic pricing models backed by a relational database. Integrated Google Maps API for optimized geospatial station discovery and reduced mobile data load latency.
Technologies: Spring Boot, JWT, Kotlin (Android), React, Google Maps API, MySQL, AWS | *[GitHub]*

ThingsBoard IoT Chatbot *March 2026 – Present*

Engineered an AI-powered IoT chatbot that integrates ThingsBoard with OpenAI GPT to enable natural language querying of real-time device telemetry and attributes. Designed a multi-layered Spring Boot backend with smart context filtering to minimize OpenAI token consumption and 60-second caching to optimize ThingsBoard API calls. Engineered multi-turn conversation memory with MySQL persistence and role-based device scoping via custom X-TB token authentication.
Technologies: Spring Boot, OpenAI API, ThingsBoard REST API, Timescale DB, RabbitMQ, Radis | *[GitHub]*

EV Station Finder *2025 – Present*

Built a full-stack EV station discovery platform combining a Spring Boot backend with a Kotlin/Jetpack Compose Android app. The backend aggregates live data from the OpenChargeMap (OCM) API and caches spatial viewport queries in Upstash Serverless

Redis (1-hour TTL) to minimize database load. Implemented Haversine-based distance scoring, spatial bounding box queries on PostgreSQL, and a daily cron sync job for OCM data freshness. The Android client follows the MVVM pattern with 1.5-second camera debounce to prevent API flooding, parallel coroutine pre-fetching for station detail cards, and SharedPreferences-backed bookmark management. Key features include connector-type filtering (CCS2, Type 2, CHAdeMO), real-time slot availability, carousel map browsing with card snapping, user reviews and ratings, and an expressway corridor route planner.
Technologies: Spring Boot, PostgreSQL (Supabase), Upstash Redis, Kotlin (Jetpack Compose), Google Maps SDK, Retrofit, OpenChargeMap API | *[GitHub]*

Certifications

- **Claude Code in Action** | Anthropic Academy | View Certificate *March 2026*
- **Software Engineering Job Simulation (Java Spring Boot)** | Hewlett Packard Enterprise | View Certificate *May 2025*
- **Core Java** | HackerRank | View Certificate *November 2024*

Achievements

- **1st Place in Binary Battleground 2K25** | Brainware University | View Certificate *February 2025*

Hackathons & Events

Smart India Hackathon 2025 | *Internal Hackathon Qualifier* | Brainware University *October 2025*

Qualified for the internal round of Smart India Hackathon 2025 as part of a 6-member team, building a civic issue reporting and management platform for Jharkhand. Solely responsible for designing and developing the entire Spring Boot backend, including REST API architecture, database design, and server-side business logic.
Technologies: Spring Boot, MySQL, REST API, HTML/CSS/JavaScript | *[GitHub]*

Languages Known

English, Hindi, Bengali

Hobbies & Interests

Riding Bike, Travelling, Solving Rubik's Cube